

LESSON GUIDE

Technology Plus

Unit 6: Database

5.4.1 – Data Backups

Lesson Overview:

In this lesson, students will explore the importance of data backups, the types of data backups, and how the storage location of a backup impacts data recovery.

Students will:

- Explore the basic concepts of data backups.
- Understand the different types of backups and their purposes.
- Recognize the importance of storage location and how each type is beneficial.

Guiding Question: Why are data backups important, what are the different types of data backups, and how is each storage location valuable?

Suggested Grade Levels: 8-12

Technology Needed: Yes

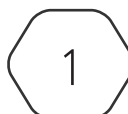
Approximate Time Required: 70-110 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 5.4 – Explain basic data backup concepts.
 - Data
 - File backups
 - System backups
 - Restoring data
 - Location
 - Stored locally
 - Flash drive
 - External hard drive
 - Secure digital (SD) card
 - Cloud storage

This content is based upon work supported by the US Department of Homeland Security's Cybersecurity & Infrastructure Security Agency under the Cybersecurity Education Training and Assistance Program (CETAP).



CYBER.ORG
THE ACADEMIC INITIATIVE OF THE CYBER INNOVATION CENTER

Copyright © 2025 Cyber Innovation Center
All Rights Reserved. Not for Distribution.

Lesson Background

Background Information:

Students will explore how to protect data by creating copies of it through backups to prevent loss due to disasters, corruption, or accidental deletion. Students will differentiate between a file and system backup, and how data that is lost can be recovered or restored from backups. Students will differentiate between local and cloud storage, explain how each storage location is valuable, and decide how to choose location type based on data needs. Students previously discussed cloud and local storage in lesson 5.2.1 – Database Fundamentals. This lesson gives more detail regarding data backup and recovery, naming common local storage devices and common cloud storage providers.

In the lab, students will explore the backup, deletion, and recovery of files when using both the CLI and GUI in the Ubuntu and Windows Server machines. The lab allows students to differentiate how the two operating systems handle file backup and recovery while also practicing the use of both CLI and GUI when interacting with computers and software.

Note: backup = noun, back up = verb

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 5.4.1 – Data Backups • Lab Slides 5.4.1 – Backup and Recovery • Activity Slides 5.4.1 – Backup Recovery Plan
Materials per Student	• Student Notes 5.4.1 – Data Backups • Student Handout 5.4.1 – Data Backups • Assessment 5.4.1 – Data Backups (optional)

Lesson Background

Cyber Terms:

Term	Definition
data backup	the process of creating copies of data to protect from loss due to disasters, corruption, or accidental deletion
file backup	backs up one or more files on a computer
system backup	backs up an entire computer system
data recovery	the process of restoring data from a backup in the event of accidental deletion, data corruption, or hardware failure; also data restoration
location	the specific place where data is physically or virtually stored
local storage	stores data on physical devices that are directly connected to a network or computer
cloud storage	stores data on remote servers managed by third party providers, accessed via the internet

5.4.1 – Data Backups

Guiding Question: Why are data backups important, what are the different types of data backups, and how is each storage location valuable?

Lesson Launch (5-10 minutes)

Don't Lose It – Use It!

- **Scenario:** Imagine you ran out of space in your cloud storage. You decide to delete a few folders to clean up space. You accidentally deleted a folder that contained files for a group project that was shared with others. Because the folder was not backed up the entire group lost access to the folder, and they must redo the work.
- **Ask:** How do you think the students feel knowing all their work has been erased?
- If the folder was backed up, the group could easily recover their files and continue their project. Regular backups are important because they allow users to protect critical data that could cause a negative impact if it is lost or corrupted. The loss or corruption of critical data can cause financial losses, legal issues, and damage to reputation, especially in business settings.
- Remind students they have learned about data and how it can be a personal and business asset. Protecting data through backups is necessary for securing its availability and integrity.

What is a data backup?

- Introduce **data backups** and their importance.
 - Remind students of management and protection strategies to prevent data loss - discussed in lesson 5.1.1 – Data.
 - Security Measures: Implement robust security protocols, including encryption, access controls, and regular security audits.
 - Backup and Recovery: Maintain regular backups and have a disaster recovery plan in place to minimize the impact of data loss or corruption.
 - Compliance: Enforce the legal, regulatory, and industry standards that apply to how an organization operates.
 - This lesson specifically focuses on backup and recovery.

Backups – A Digital Lifeboat (25-35 minutes)

- Distribute Student Handout 5.4.1 – Data Backups and encourage students to fill in the answers during the lesson.

File Backups

- Introduce **file backups** and the types of file backups.
- Discuss the purpose of file backups and how they can be used to recover data.
- Depending on what applications or software your school uses, take this time to show students how to restore a file using version history.
- Real-life Example: Jane was working on a meaningful report for work. One day she came to work and was notified of an attack against the company that caused all their data to go missing. Many employees were not worried because they backed up their files. Jane, along with a few others, had not backed up any files. Now she must recreate her report from memory, causing a delay in business processes. Jane is frustrated and devastated, but she has no other options.

System Backups

- Introduce **system backups** and the types of system backups.
- Discuss the purpose of system backups and how they can be used to recover operating systems.
- Depending on what device your school uses, take this time to show students the backup and recovery options found in their device settings.
- Real-life Example: Bill works for a marketing company and received an email that contained an attached invoice. Bill opened the attachment and unknowingly downloaded a virus to his computer and company network. The virus quickly spread, and all company files were deleted across computers and servers. The company did not have any backups and only a few employees saved files on their local computers. Ninety percent of data was lost and there was no way to recover it. Employees could no longer work, the business lost clients, and their reputation suffered.

Restoring Data/Data Recovery

- Introduce data restoration, also known as **data recovery**.
- Discuss data recovery in relation to file and system backups.
- Mention that data recovery can require specific backup software or tools to be installed on their device.

Data Recovery Plan

- Mention that everyone with critical data should have a recovery plan to be prepared for any unexpected data loss.
- Discuss the five questions that can help one prepare a data recovery plan.
- Students will create their own data recovery plan in an activity at the end of the lesson.

Location

- Discuss storage location and how it affects data backup and recovery.
- Students will explore both local and cloud storage locations, and what they should consider before choosing a storage method.
- Unit 3: Hardware goes more in-depth about storage devices and storage units.

Local Storage

- Discuss **local storage** and common types of local storage.
 - As you discuss the types of local storage devices, if time allows, show students images of the varying types of flash drives and external hard drive connectors.
- Discuss **flash drives**, their capacity range, and their most common sizes.
 - Common USB ports are USB-A and USB-C.
- Flash Drive Advantages:
 - Portability: Easy to transport and use with different devices.
 - Ease of Use: Simple plug-and-play (PnP) functionality.
 - Plug and Play is when users can plug in a device and the system will recognize and configure the device without any additional steps.
 - Affordability: Cost-effective for lower capacities.
- Flash Drive Disadvantages:
 - Durability: Can be easily lost or damaged.
 - Capacity: Limited compared to external hard drives.
 - Speed: Slow read and write speeds compared to larger storage options.
- Discuss **external hard drives**, their capacity range, and their most common sizes.
 - A Thunderbolt connector utilizes the USB-C connector and can transfer data at higher speeds than a traditional USB-C.
 - An eSATA connector, not used as often, is faster than a USB but slower than Thunderbolt.
- External Hard Drive Advantages:
 - High Capacity: Suitable for large backups or extensive data storage.
 - Performance: Fast read and write speeds compared to flash drives and SD cards.
 - Durability: Sturdy for frequent use.
- External Hard Drive Disadvantages:
 - Portability: Large and can be inconvenient to carry around.
 - Cost: More expensive than flash drives and SD cards, especially for high-capacity models.
- Discuss **secure digital (SD) cards**, their capacity, and their most common sizes.
- Mention there is a difference between SD cards and solid-state drives (SSDs), students learn about SSDs in lesson 2.3.1 – Storage Devices.
- SD Card Advantages:
 - Portability: Compact, easy to carry and use in devices with SD card slots.
 - Versatility: Compatible with many consumer electronics and computers.
- SD Card Disadvantages:

- Speed: Slow read and write speeds when compared to external hard drives.
- Durability: Can be easily lost or damaged.
- Cost: High-capacity models can be expensive.

Cloud Storage

- Discuss **cloud storage**, its capacity, convenience in comparison to local storage, advantages, and disadvantages.
 - This lesson refers to cloud storage as a public model, but more information in Lesson 2.6.1 – Cloud Technologies.
- Cloud Storage Advantages:
 - Accessibility: Accessible from any location with internet connectivity.
 - Scalability: Storage capacity easily adjusts based on needs.
 - Backup and Recovery: Automated backups and data recovery options.
 - Collaboration: Easy sharing and collaboration features.
- Cloud Storage Disadvantages:
 - Internet Dependency: Requires a stable internet connection for access.
 - Outgoing Costs: Subscription fees can accumulate over time and increase with growing storage needs.
 - Security Risks: Potential for data breaches, even with robust security measures.
 - Limited Control: Dependence on the provider for data management and recovery.
 - Time Consuming: Backing up and restoring data from the cloud can take longer compared to using a local storage device.
- Discuss the common providers of cloud storage for personal and business use.
 - Free cloud storage is great for personal use, paid cloud storage can still be used for personal reasons.
 - Paid cloud storage is most often used by businesses. AWS offers a pay-as-you-go model. Microsoft SharePoint is OneDrive for Business.
 - All cloud services can be accessed using an internet browser or an app using smart devices.

Lab: 5.4.1 Backup and Recovery (30-45 minutes)

- This lab requires access to the CYBER.ORG Range, the Ubuntu Linux machine, the Windows Server 22 machine, and Lab Slides 5.4.1 – Backup and Recovery.
- Introduce the guiding question and objectives for the lab.
- Students will use the Ubuntu and Windows VMs to backup, delete, and recover files using GUI and CLI interfaces.
- Discuss lab key terms.
- Students should open two separate tabs in their browser to launch each machine. This will allow them

to easily switch back and forth during the lab.

- Students should complete the lab portion of their Student Handout.

Helpful Tips

- Directories and folders are essentially the same with “folder” being the GUI metaphor used to represent a directory.
- Type carefully when using CLI because a typo can copy files to the wrong places or give folders the wrong name.
- Remind students that permissions were covered in lesson 1.2.3 – Octal and 3.1.2 – File System Management.
- Ubuntu: `~` in front of a /path navigates to the home directory, `ls` command shows the contents in a folder, `-lh` command shows the size, modification date and time, and the name of files in a folder.
- Windows: `dir` shows all the files in a directory.
- Both Ubuntu and Windows use `mkdir` command in CLI for directory/folder creation.
- Using a slash (/) at the end of a source when copying files with the CLI will ensure the contents of the folder are copied and not just the folder.
- Windows: The **robocopy** utility copies file data from one location to another.
- Ubuntu: The **rsync** utility creates incremental copies of files. `-avh` will preserve the structure, show the name, and the size of the files.
- Windows: If you do not hold SHIFT while copying files, they will not copy to the clipboard and cannot be pasted to their destination.
- “Copy to” and “Move to” are different GUI functions. “Copy to” should be used to backup files instead of “Move to”. “Move to” will only move the files to the destination location.
- “Copy” and “Cut” are different GUI functions. “Copy” should be used to backup files instead of “Cut”. “Cut” will delete the files from the source location and paste them to the destination location.
- Windows: `rmdir` will delete a directory or folder. `/S` will delete all files and subfolders in the directory or folder.
- Ubuntu: `rm` is the remove command and `-r` specifies to delete a directory or folder and its contents.

Put It All Together

- What is the difference between the **robocopy** and **rsync** commands?

Robocopy is used in Windows CLI and creates copies of files from one location to another. Rsync is used in Linux CLI and creates incremental copies of files.

- Is it easier to use CLI or GUI to backup and recover files?

It is easier to use GUI if you are not familiar with CLI. CLI has a learning curve while GUI allows a user to interact with the computer using windows, buttons, and icons.

- Would CLI or GUI be more beneficial for backup and recovery of multiple data at once?

CLI is more beneficial for backup and recovery of multiple data at once because it is easier to type commands in the terminal to backup or restore multiple files or folders at once, instead of copying and pasting each piece of data one file or folder at a time.

- Did the Ubuntu and Windows machines have any similarities in backup and recovery?

Both machines allow users to backup and recover files using CLI and GUI. Both machines use the command mkdir to create a directory or folder. Both machines provide a pop-up if a user is trying to place duplicate files in the same folder or directory. Both machines use Source/path Destination/path in the CLI to backup or recover files.

Activity: Data Recovery Plan (10-15 minutes)

- Students will create their own data recovery plan.
- This plan should be realistic and something students can and would follow to protect their data from accidental loss.
- Activity Slides 5.4.1 has guiding questions that will help students complete their data recovery plan on the Student Handout.
- Encourage students to complete the plan independently so it is specific to their needs.

1. What to backup

Students should list any device and specific data they want to backup, i.e. Phone - photo album.

2. Where to backup

Students should choose one of the following and specify what device or providers they would use based on today's lesson. Cloud storage – Google Drive, OneDrive, or iCloud, or any other common cloud provider for personal usage. Local storage – flash drive, external hard drive, SD card, or any other type of local storage device. i.e. Cloud storage – OneDrive Local Storage – flash drive.

3. Backup Schedule

Students should write daily, weekly, or monthly and describe why, i.e. I will backup monthly because I only take a few photos a month.

4. Recovery Plan

Students should write how they will recover any data loss based on the device they used, the data backed up, and where the backup is stored, i.e. If a photo is accidentally deleted or missing, I will sign into OneDrive and download the recovery photo to my phone.

5. Recovery Test

Students should write their recovery test plan, i.e. I will take a random photo, backup the photo to OneDrive, delete the photo from my phone, log into OneDrive, and recover the photo.

Putting It All Together (5-10 minutes)

- Review what students learned today by discussing the questions below as a class.
- Why is it important to back up data?

Data backups can prevent the loss of data in the event of disasters, system failure, or accidental deletion. Backups can be recovered and aid business continuity, enforce compliance, and ensure data availability and integrity.

- What is the difference between a file backup and a system backup?

A file backup focuses on saving copies of individual files or folders. A system backup will save a copy of an entire OS including programs, applications, and personal data.

- How does storage location affect data backups and what should be considered before choosing a location?

Storage location affects the reliability, accessibility, and integrity of a data backup. A user should consider the characteristics, capacity, advantages, and disadvantages of each storage option before deciding.

- What are the common types of local storage?

Common types of local storage include flash drives, external hard drives, and SD cards.

- What are the benefits of cloud storage?

The benefits of cloud storage are storing data off-site allowing for easy recovery in the event of deletion at the on-site location, scalability as needs shift, and accessibility from anywhere increasing collaboration.